

ROUNAK SINGH

Data & ML Engineer | Generative AI & Infrastructure

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PROFESSIONAL SUMMARY

Results-oriented IT professional with over 5.5 years of combined experience across production data engineering, cloud architecture, MLOps, and applied machine learning. Proven track record of architecting scalable AWS and Snowflake pipelines handling 500K+ daily records for Fortune 100 clients, alongside automating complex CI/CD frameworks. Extensive US-based research experience and a German market exposure, currently specializing in production-grade Enterprise Generative AI and scalable Retrieval-Augmented Generation (RAG) architectures.

CORE SKILLS

GenAI & NLP:	LLM Orchestration, Vector DBs (Qdrant, Pinecone), Prompt Engineering, Semantic Chunking, Whisper API, BERT, HuggingFace.
Data Engineering:	Apache Airflow, AWS, Apache Spark, Snowflake, dbt, ETL/ELT pipelines, Azure, GCP.
MLOps & DevOps:	Docker, Kubernetes, GitLab CI/CD, Jenkins, Terraform, AWS SageMaker, Data Drift & Model Stability Monitoring.
Languages & DBs:	Python, SQL, MySQL, PostgreSQL, DynamoDB.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Hexaware Technologies Inc., USA Feb 2022 – Feb 2025

- Built Enterprise data lake for a Fortune 100 client providing vetted & federated access of upstream data.
- Owned and maintained ETL pipelines that process 500K+ records across 40+ tables for daily uploads.
- Integrated heterogeneous data from 14+ siloed systems into unified structures, enhancing data availability for downstream analytics and ML based prediction applications.
- Developed robust Apache Airflow DAGs to orchestrate complex ETL workflows, managing retries, dependencies, and automated alerting to boost system uptime by 35%.
- Built automated CI/CD infrastructure pipelines using GitLab, Terraform, and Jenkins, successfully reducing software deployment lifecycles by 40%.
- Deployed MLOps practices to systematically monitor feature distribution, data drift, and production model stability by 15%.
- Successfully modernized legacy infrastructures by managing structural data migrations from IBM Netezza and Talend over to Snowflake and AWS.

Research Assistant (Machine Learning) | University of Missouri, USA Jun 2020 – Dec 2021

- Designed and trained specialized Machine Learning and Deep Reinforcement Learning models (A3C, Actor-Critic networks) for autonomous drone path finding optimization.

- Built data preprocessing, automated feature engineering, and inference pipelines to systematically eliminate data leakage and improve inference accuracy.
- Processed and engineered 120,000+ geospatial data records to simulate dynamic environments and evaluate last-mile logistics use cases.
- Published and Peer Reviewed Applied ML research articles across various International Conferences.

PROFESSIONAL UPSKILLING & INTERNATIONAL MARKET RESEARCH

Applied AI Upskilling | India

Feb 2025 – Present

- Following completion of US work engagement, took a deliberate career investment break to transition from data engineering into applied Generative AI.
- Pursuing Data Science & Machine Learning Program at Scaler (remotely) covering advanced ML, deep learning, and production grade ML systems architectures.
- Independently researched and designed AI systems providing solutions to real-world problems I encountered (details in portfolio).

Independent AI Market Exploration | Germany

Feb 2026 – May 2026

- Moved to Germany by securing the Chancenkarte (Opportunity Card) to explore market in AI related research and industry opportunities across EU.
- To understand industry wide AI transition, gained knowledge of GDPR-compliant deployment patterns and EU AI Act compliance frameworks applicable to enterprise AI rollouts.

PROJECT PORTFOLIO

Enterprise Multimodal RAG Platform | *OpenAI, LlamaIndex, AWS, Qdrant, Docker*

- Built a scalable RAG platform supporting document, audio and structured data ingestion & retrieval.
- Implemented semantic chunking, vector search, hybrid retrieval and reranking pipelines.

Multilingual OCR & Intelligent Translation | *Microsoft TrOCR, NLP, Translation systems*

- Built AI based translator by reverse-engineering image identification and machine translation.
- Developed an automated pipeline capable of extracting, classifying, and translating text embedded within images without manual intervention.

LLM-Based Clinical Named Entity Recognition | *GPT-4, React, FastAPI, Whisper, Docker, FHIR API*

- Engineered an AI system that ingests dictated clinical text and identifies oncological anomalies and flags severe health risks.

Multi-Agent Drone Optimization | *Deep Reinforcement Learning, TensorFlow, Python*

- Implemented Actor-Critic models to optimize routing to solve disaster-response use-cases.
- Formulated Markov-Decision Process (MDP), state-action pairs and rewards for test-based simulations for multi-packet transfer outperforming dynamic programming-based methods by ~30%.

EDUCATION

Data Science & Machine Learning Program

Jul 2025 – Present

Scaler, India (Remote)

M.Sc, Electrical Engineering

Aug 2019 – Dec 2021

PUBLICATIONS (IEEE Member : 100474300)

I. Journal Papers, Articles

1. C. Qu, F. Sorbelli, R. Singh, P. Calyam, S. Das. “Environmentally Aware and Energy-Efficient Multi-Drone Coordination and Networking for Disaster Response” To appear in IEEE Transactions on Network and Service Management (TNSM), 2023. (Impact Factor: 4.76)

II. Book Chapters

1. **R. Singh**, C. Qu, A. Esquivel Morel, P. Calyam. “Location Prediction and Trajectory Optimization in Multi UAV Application Missions” Book Chapter for ‘Intelligent Unmanned Air Vehicles for Public Safety Networks’, Book: **Springer** ‘Unmanned System Technologies’, 2021.

III. Conference Papers

1. C. Qu, **R. Singh**, Sharan Srinivas, P. Calyam, “Environmentally Aware Robotic Vehicle Networks Routing Computation for Last-mile Deliveries” IEEE International Conference on Computer Communications and Networks (ICCCN), 2023. (Invited Paper)
2. C. Qu, **R. Singh**, A. Esquivel Morel, P. Calyam. “Learning-based Multi-Drone Network Edge Orchestration for Video Analytics” IEEE International Conference on Computer Communications (INFOCOM), 2022. (Acceptance Rate: 19.8%)
3. C. Qu, **R. Singh**, A. Esquivel Morel, F. Sorbelli, P. Calyam, S. Das. “Obstacle-Aware and Energy-Efficient Multi-Drone Coordination and Networking for Disaster Response” IEEE Conf. on Network and Service Management (CNSM), 2021. (Acceptance Rate: 19.3%)
4. C. Qu, **R. Singh**, A. Esquivel Morel, P. Calyam. “Learning-based Multi-Drone Network Edge Orchestration for Video Analytics” IEEE Transactions on Network and Service Management (TNSM), 2023. (Impact Factor: 5.3)

IV. Posters, Presentations, and Demos

1. C. Qu, **R. Singh**, A. Esquivel, F. Betti, P. Calyam, and S. K. Das. “Poster: Multi-Drone Coordination and Networking Experiments for Disaster Response” NSF AERPAW Fall Event, 2021.
2. C. Qu, **R. Singh**, F. Betti, P. Calyam, and S. K. Das. “Poster: Obstacle-Aware and Energy-Efficient Multi-Drone Coordination and Networking for Disaster Response” MIZZOU Graduate Forum, 2021.